Al Hayat Hospital Network Project – Practical Project

Computer Networks – 2025 / Semester 1

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1. Introduction

This is a practical project for designing and configuring a working network for Al Hayat Hospital. The hospital has three buildings: Outpatient Clinics (Building A), Laboratories & Pharmacy (Building B), and Administration & Support (Building C). The goal was to build a secure, scalable network that supports daily hospital work, medical systems, administrative tasks, and provides separate staff and guest Wi-Fi.

Hospital Layout:

BUILDING A – OUTPATIENT CLINICS

- Reception: 4 computers + 1 network printer
- Waiting Area: Guest Wi-Fi
- 10 Consultation Rooms: 1 computer + 1 IP phone each
- 4 Administrative Offices: 2 computers each

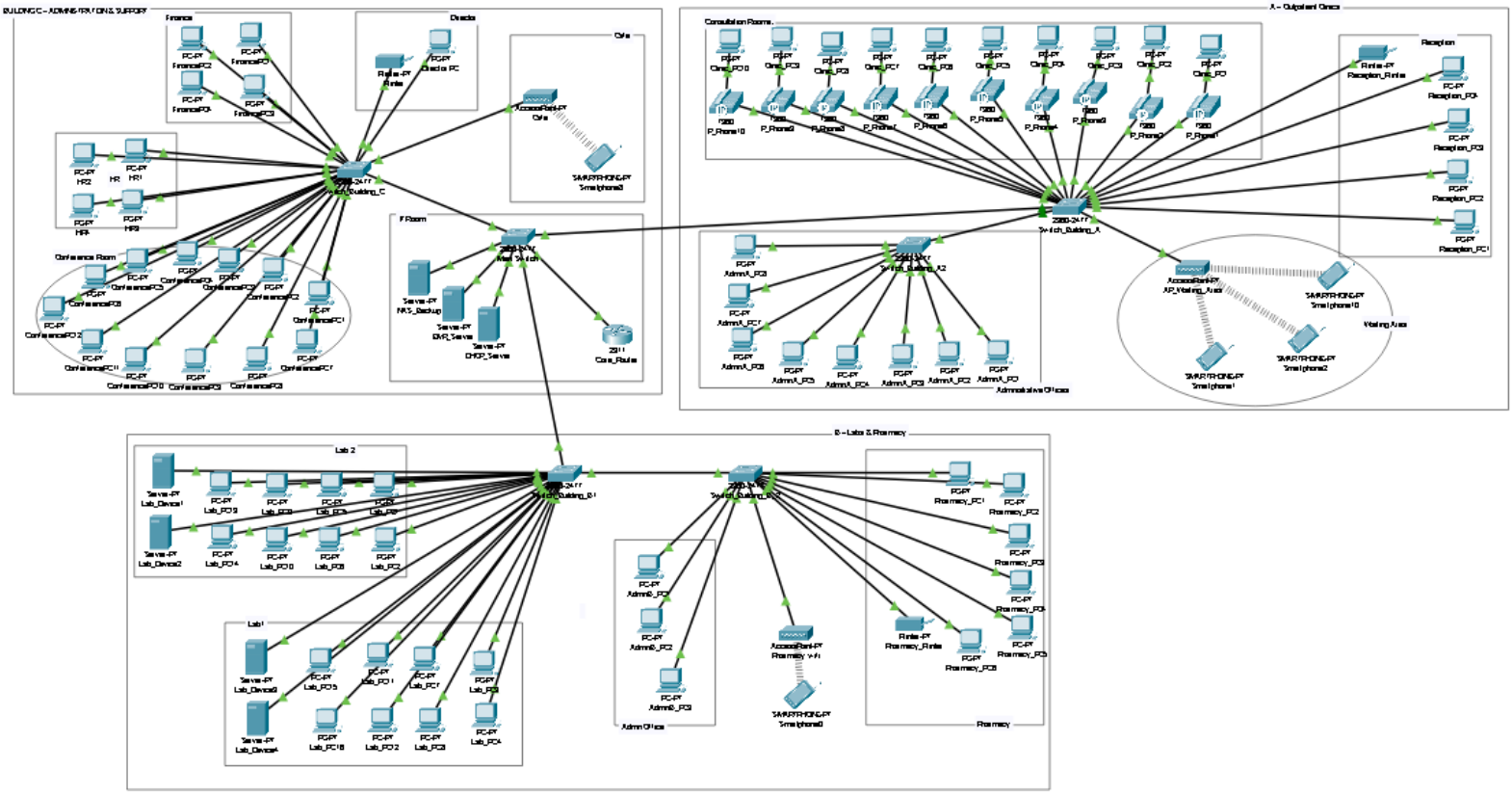
BUILDING B – LABORATORIES & PHARMACY

- Lab 1: 8 computers + 2 lab devices (Lab devices are represented by Servers)
- Lab 2: 8 computers + 2 lab devices (Lab devices are represented by Servers)
- Pharmacy: 6 computers + 1 printer
- Admin Office: 3 computers

BUILDING C – ADMINISTRATION & SUPPORT

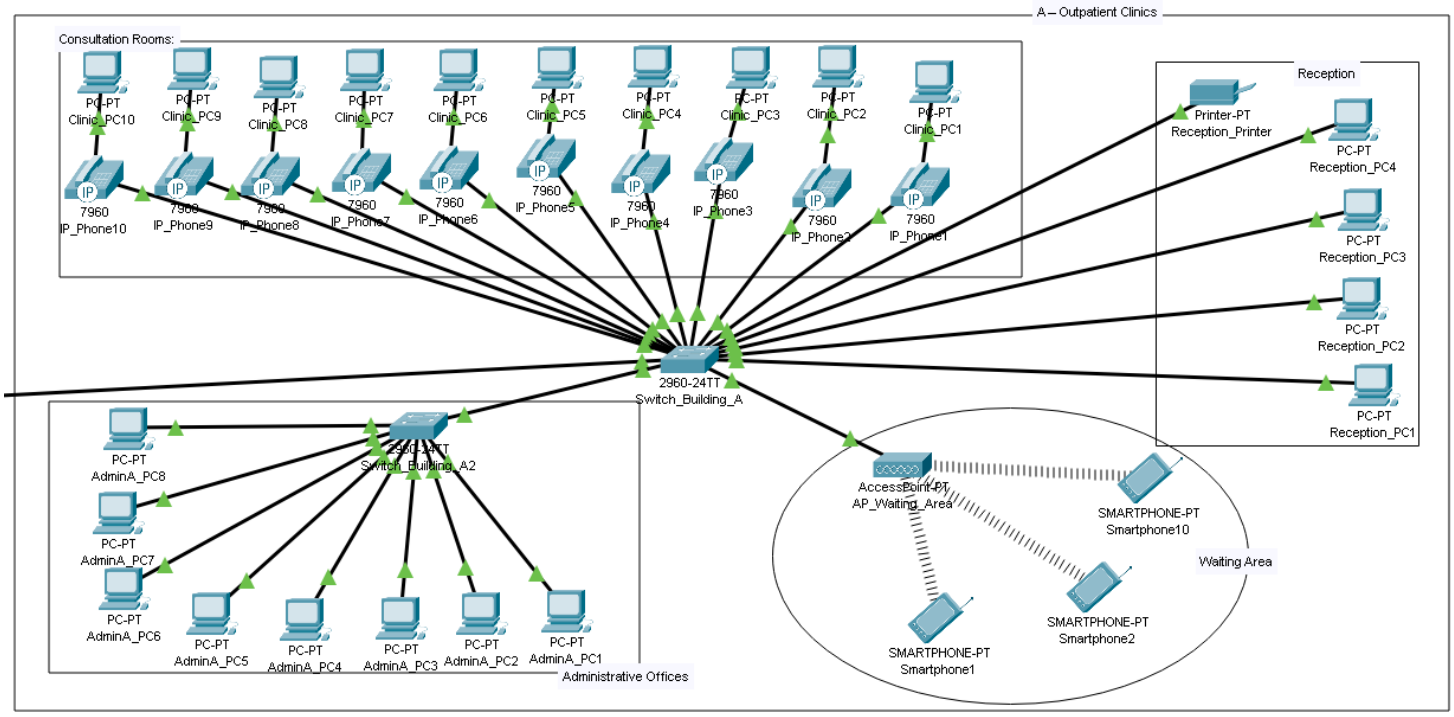
- Director: 1 computer + 1 printer
- Finance: 4 computers
- HR: 4 computers
- Conference Room: 12 computers
- Employee Café: Staff Wi-Fi
- IT Room contains:
  - Core Router
  - Main Switch
  - DHCP Server
  - EMR Server
  - Backup NAS Storage

2. Network Topology Diagram

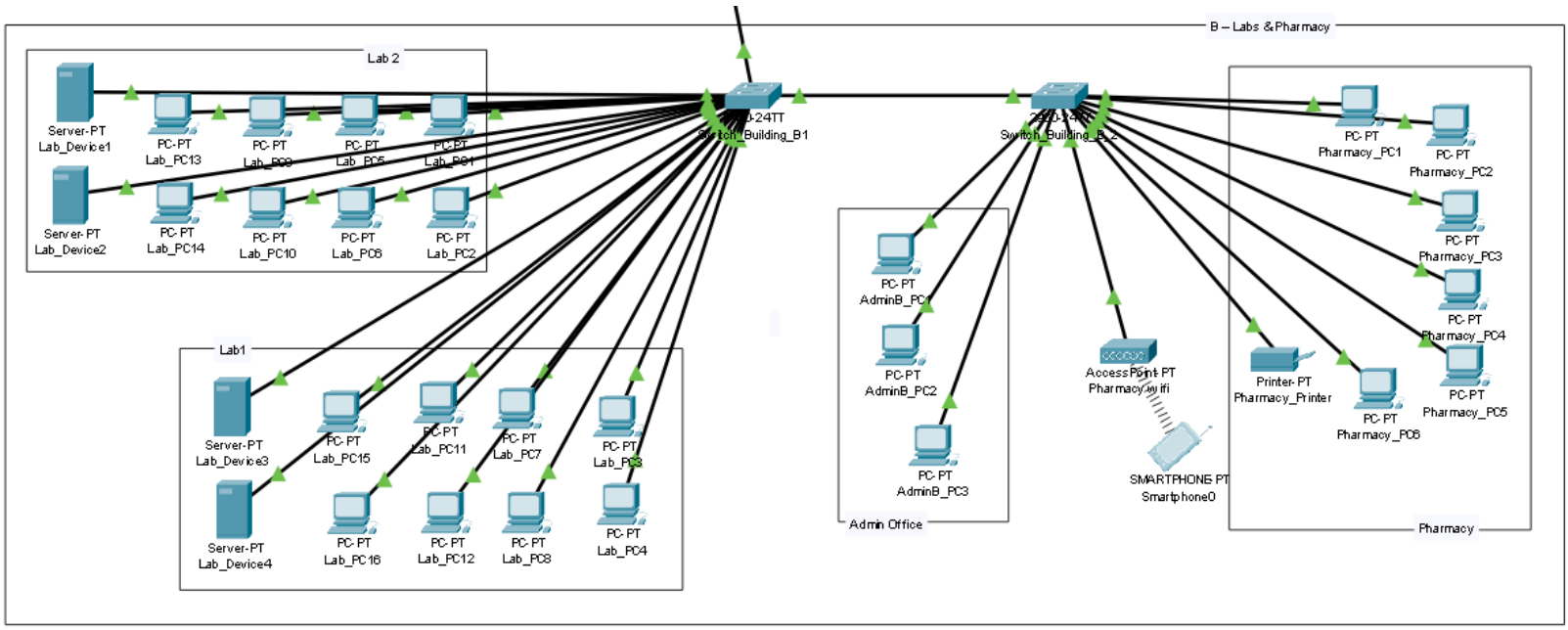


This diagram shows how switches, routers, servers, access points, PCs, and phones are connected in the Packet Tracer design.

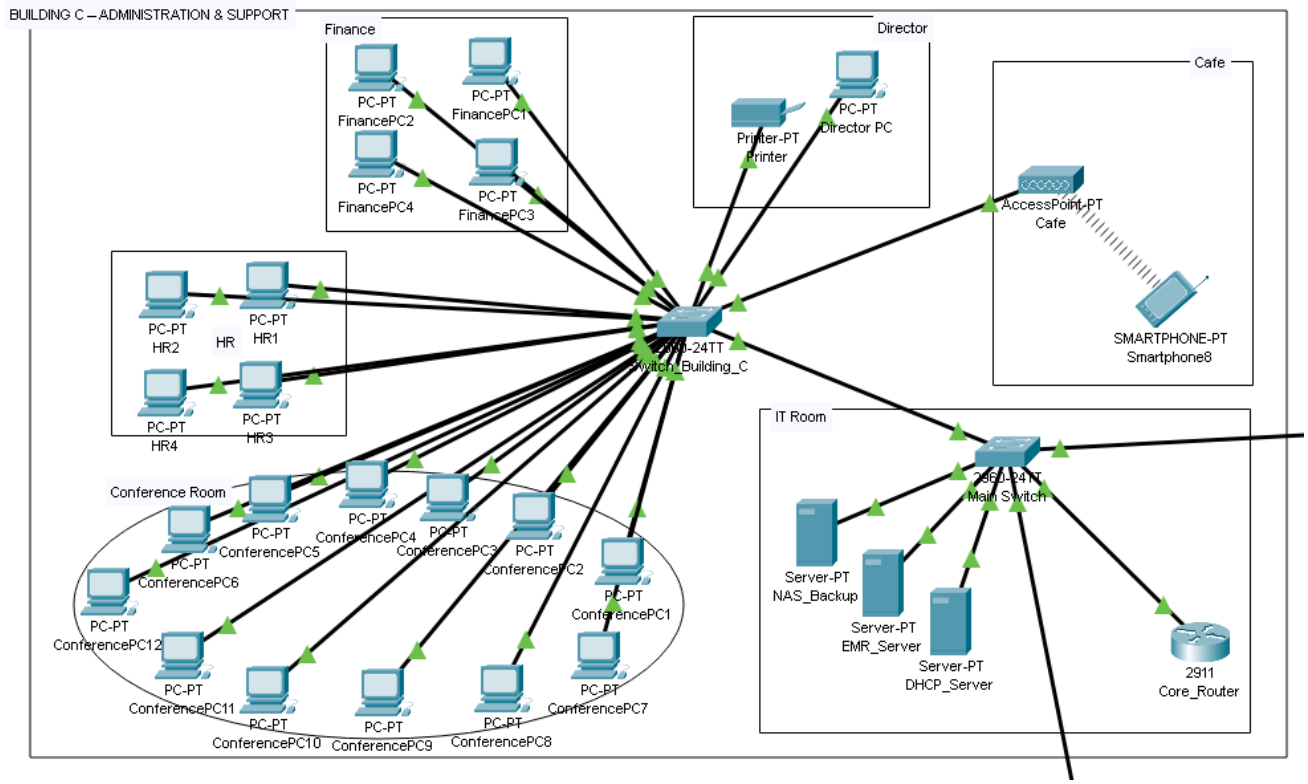
Building A Topology



Building B Topology



Building C Topology

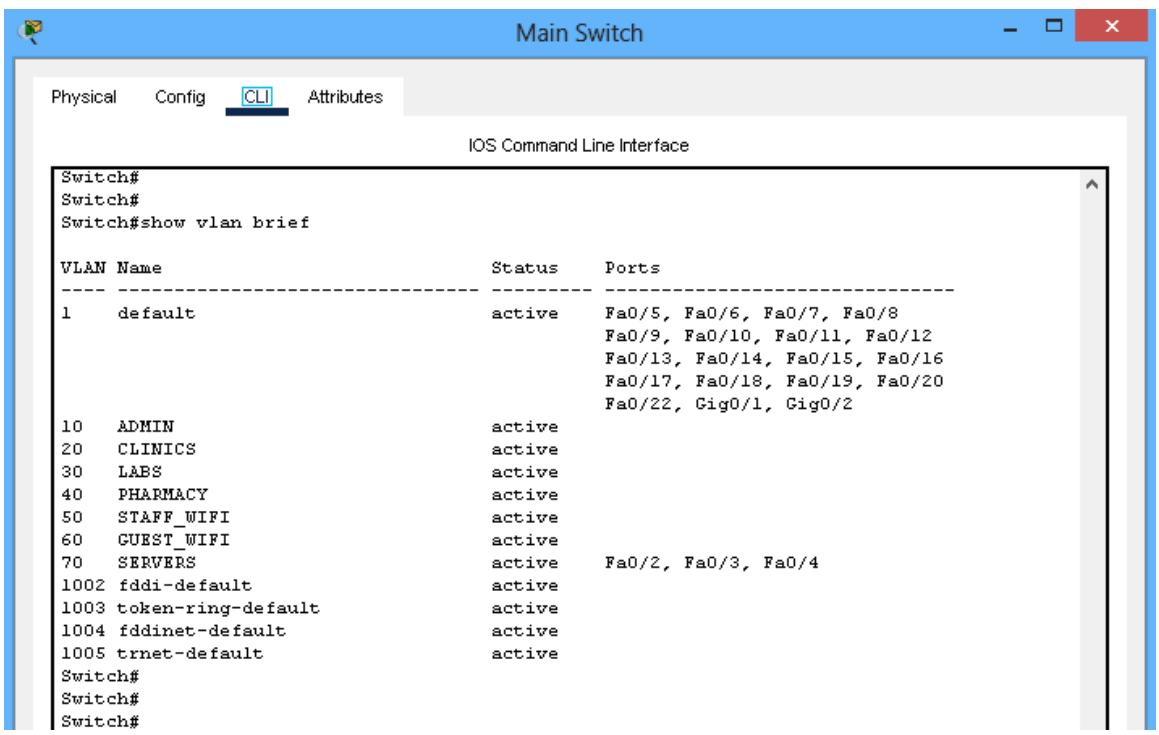


3. VLAN and IP Address Table

VLAN ID	VLAN Name	Purpose	Subnet	Gateway
10	ADMIN	Admin PCs (Director, Finance, HR)	192.168.10.0/24	192.168.10.1
20	CLINICS	Reception & consultation rooms	192.168.20.0/24	192.168.20.1
30	LABS	Laboratory systems	192.168.30.0/24	192.168.30.1
40	PHARMACY	Pharmacy computers & printer	192.168.40.0/24	192.168.40.1
50	STAFF_WIFI	Employee wireless	192.168.50.0/24	192.168.50.1
60	GUEST_WIFI	Guest wireless access	192.168.60.0/24	192.168.60.1
70	SERVERS	Servers (DHCP/EMR/NAS)	192.168.70.0/24	192.168.70.1

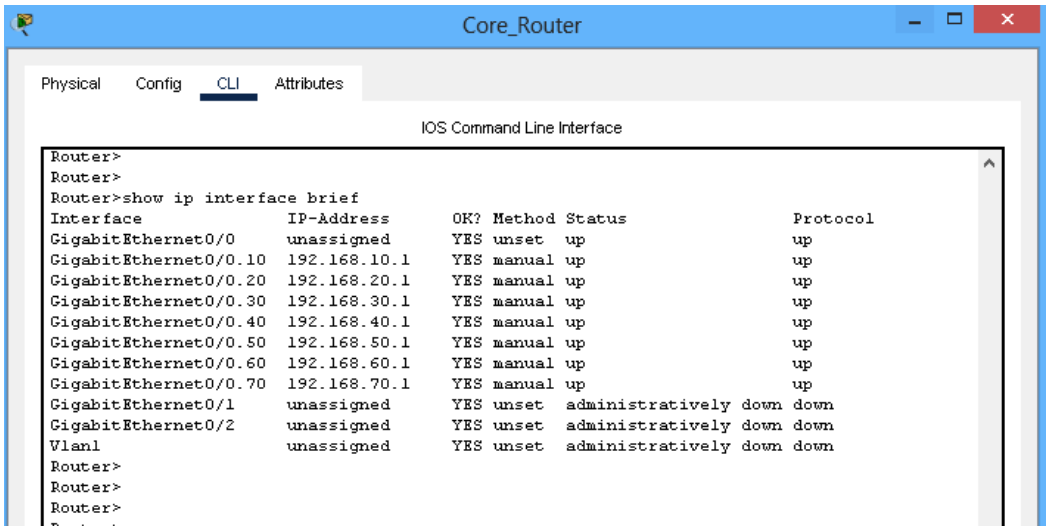
This helps each device segment stay in its own network and allows routing between them using the core router.

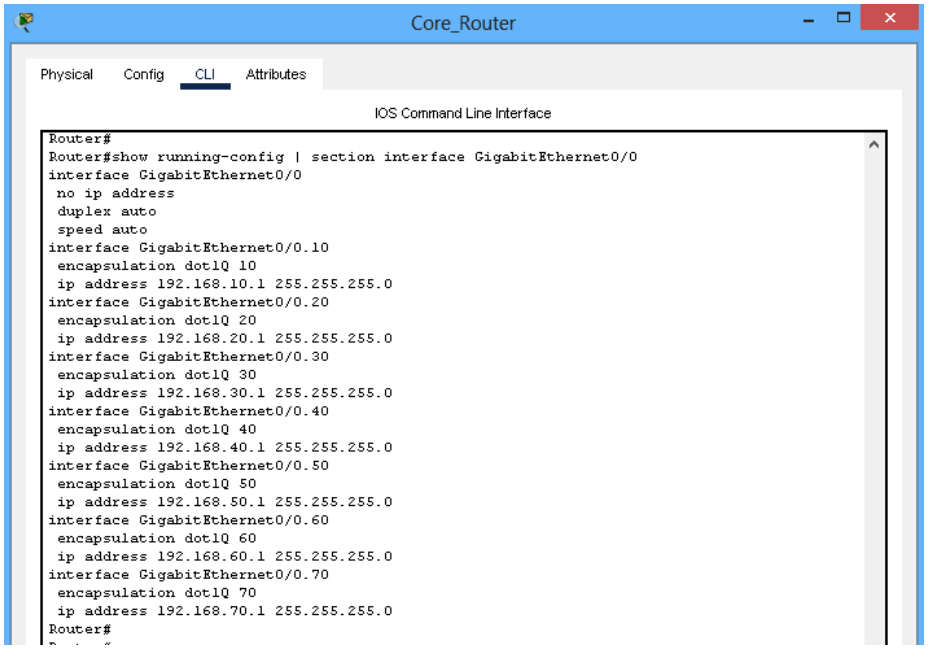
VLAN configuration on Main switch:



4. Configurations

4.1 Router Inter-VLAN Routing

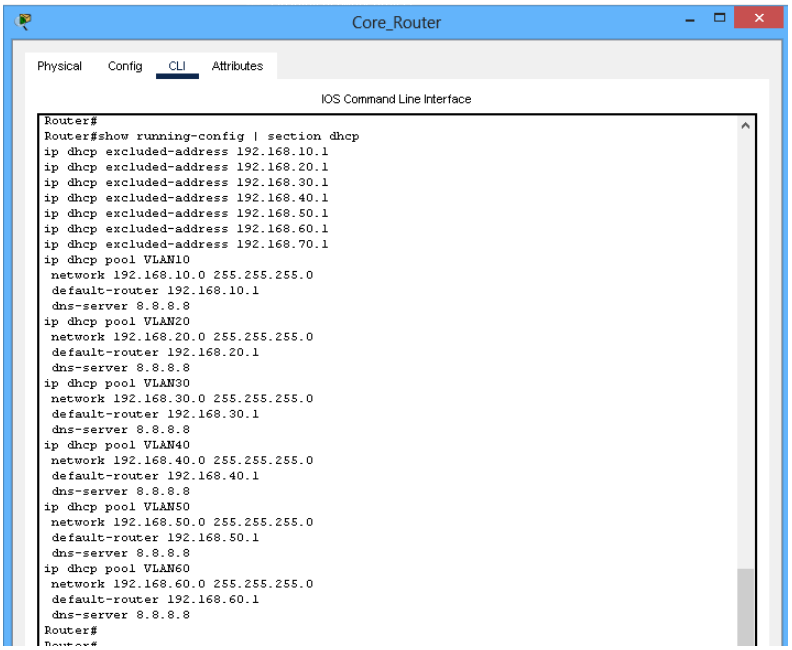




We used Router-on-a-Stick for inter-VLAN routing:

```
interface gig0/0.10
 encapsulation dot1Q 10
 ip address 192.168.10.1 255.255.255.0
...
! Repeated for VLANs 20,30,40,50,60,70
```

4.2 DHCP Server Setup



DHCP pools were created for each VLAN so devices get IPs automatically:

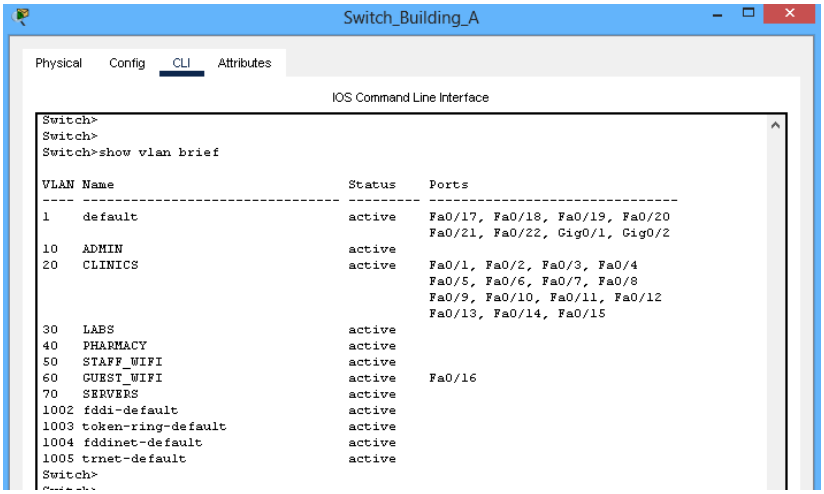
```
ip dhcp pool VLAN20
 network 192.168.20.0 255.255.255.0
 default-router 192.168.20.1
 dns-server 8.8.8.8
```

Similar commands were made for other VLANs.

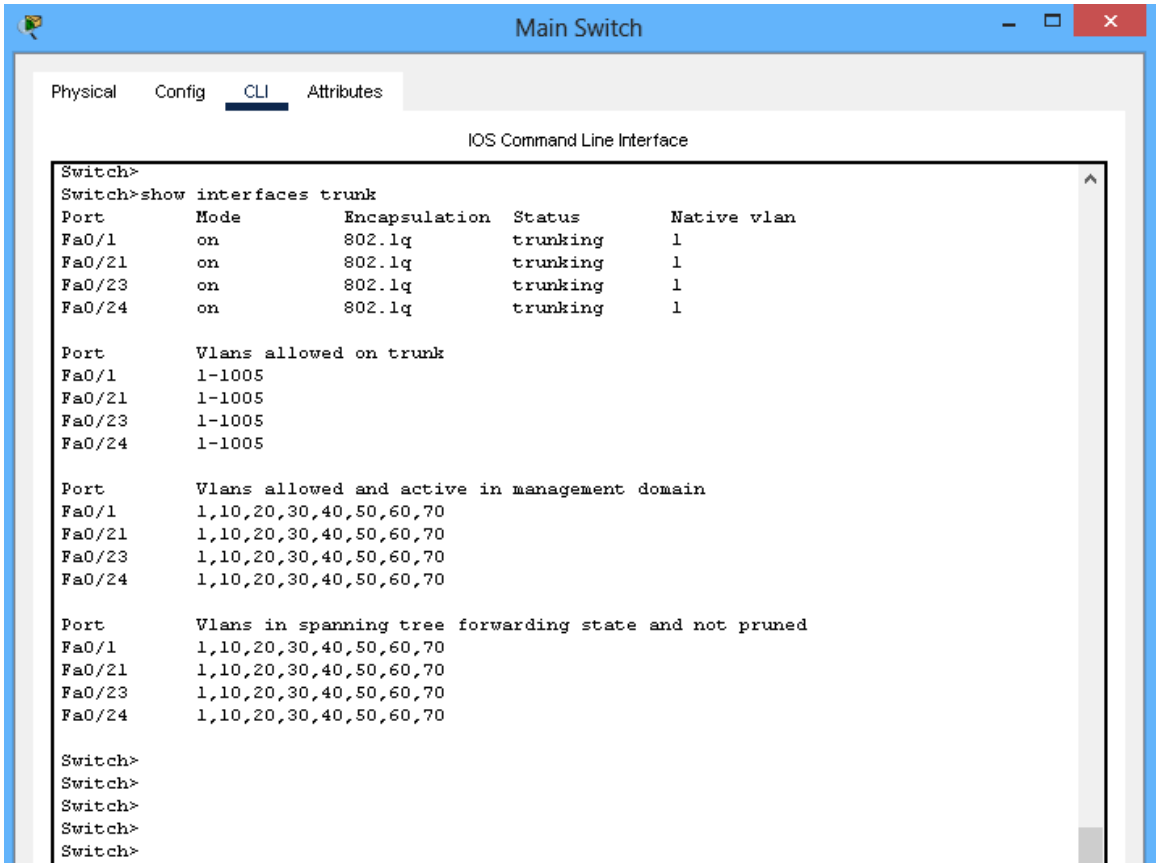
4.3 Switch VLAN & Trunk Config

Each access switch was configured so:

- End devices get assigned to correct VLANs (Admins to VLAN10, Clinics to VLAN20, etc.)



- Trunk ports carry all VLANs between switches and to the router.



Examples:

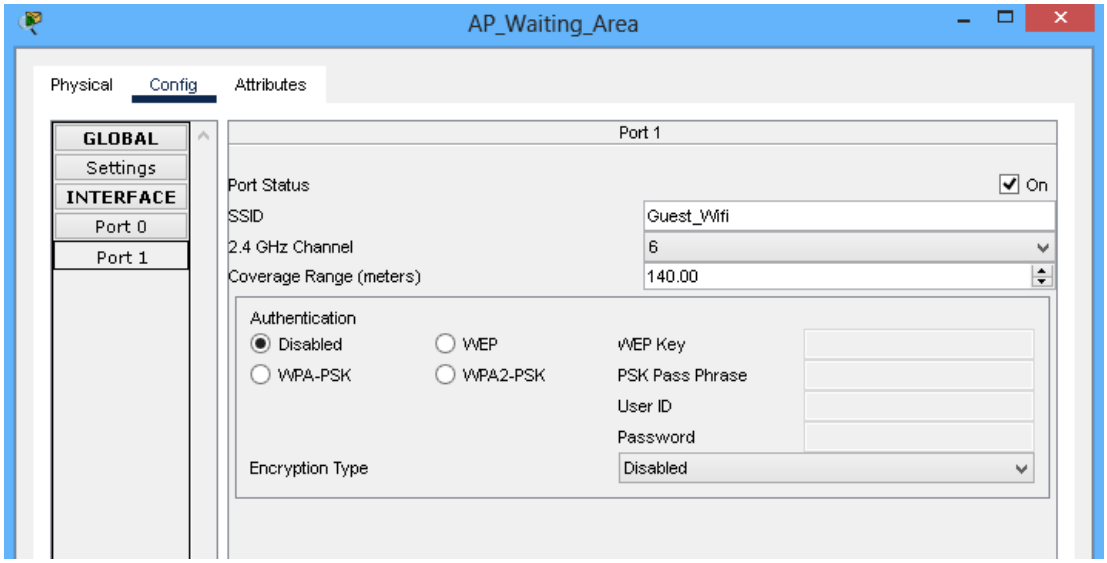
```
interface range fa0/6-15
 switchport mode access
 switchport access vlan 20
 switchport voice vlan 20
```

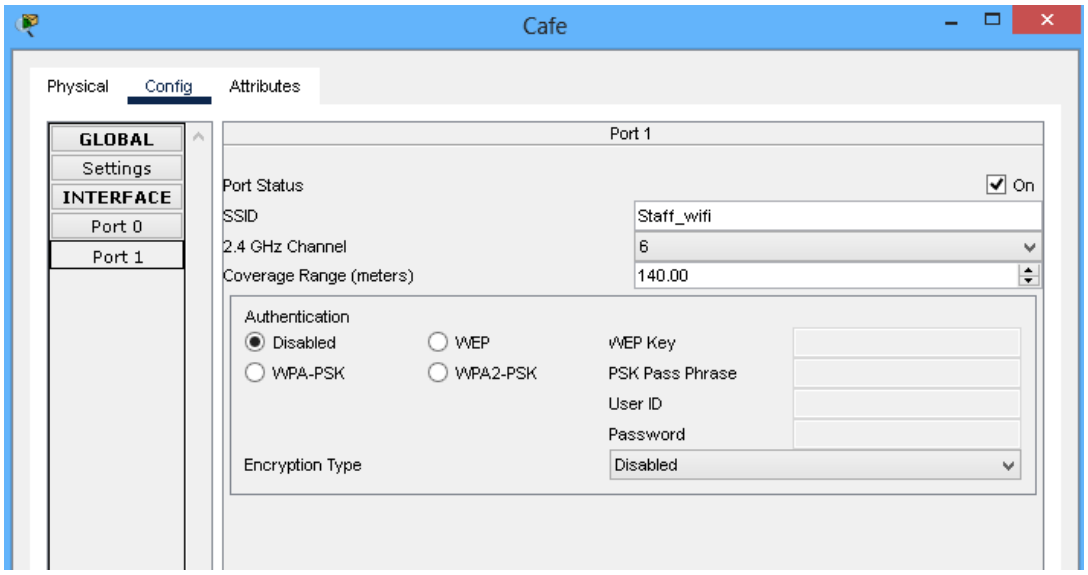
4.4 Access Points and Wi-Fi

For each AP:

- Set an SSID (e.g., Guest_WiFi, Staff_WiFi)
- The switch port the AP connects to is assigned the correct VLAN (e.g., VLAN60 for guest Wi-Fi).

Phones and mobiles connect wirelessly using the correct SSID and get IP via DHCP.

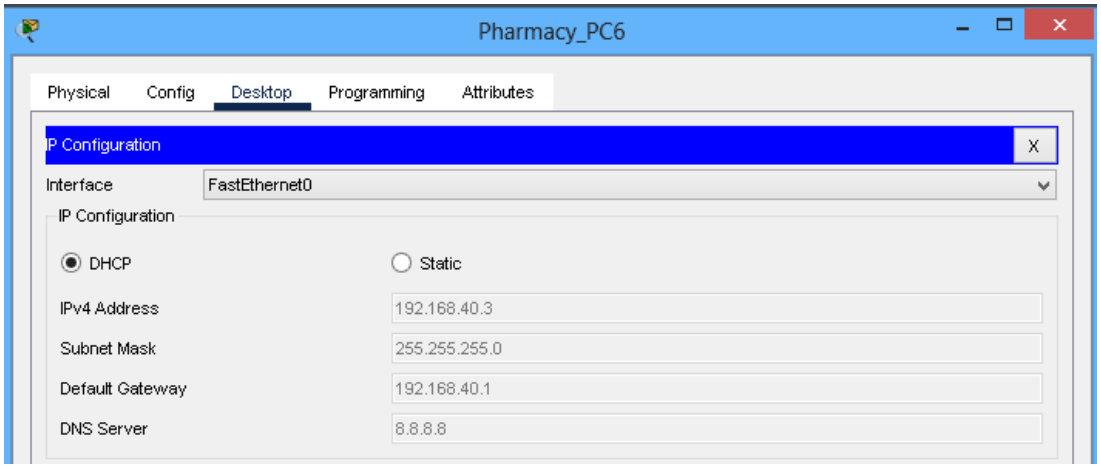


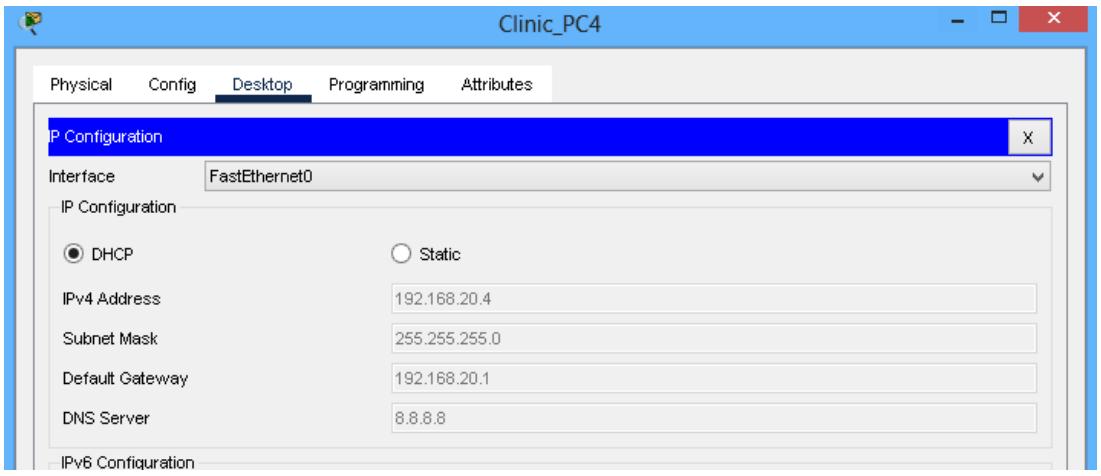


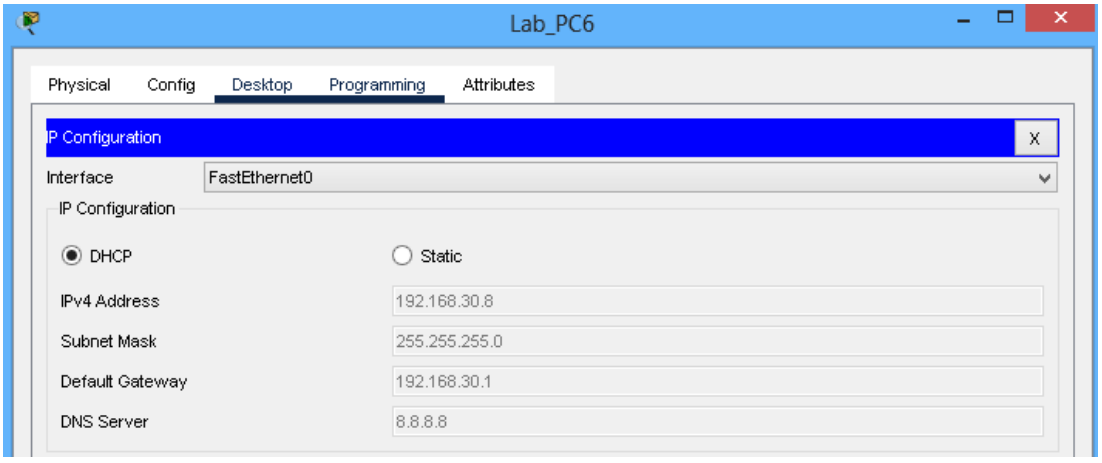
5. Testing Results

5.1 Device Connectivity

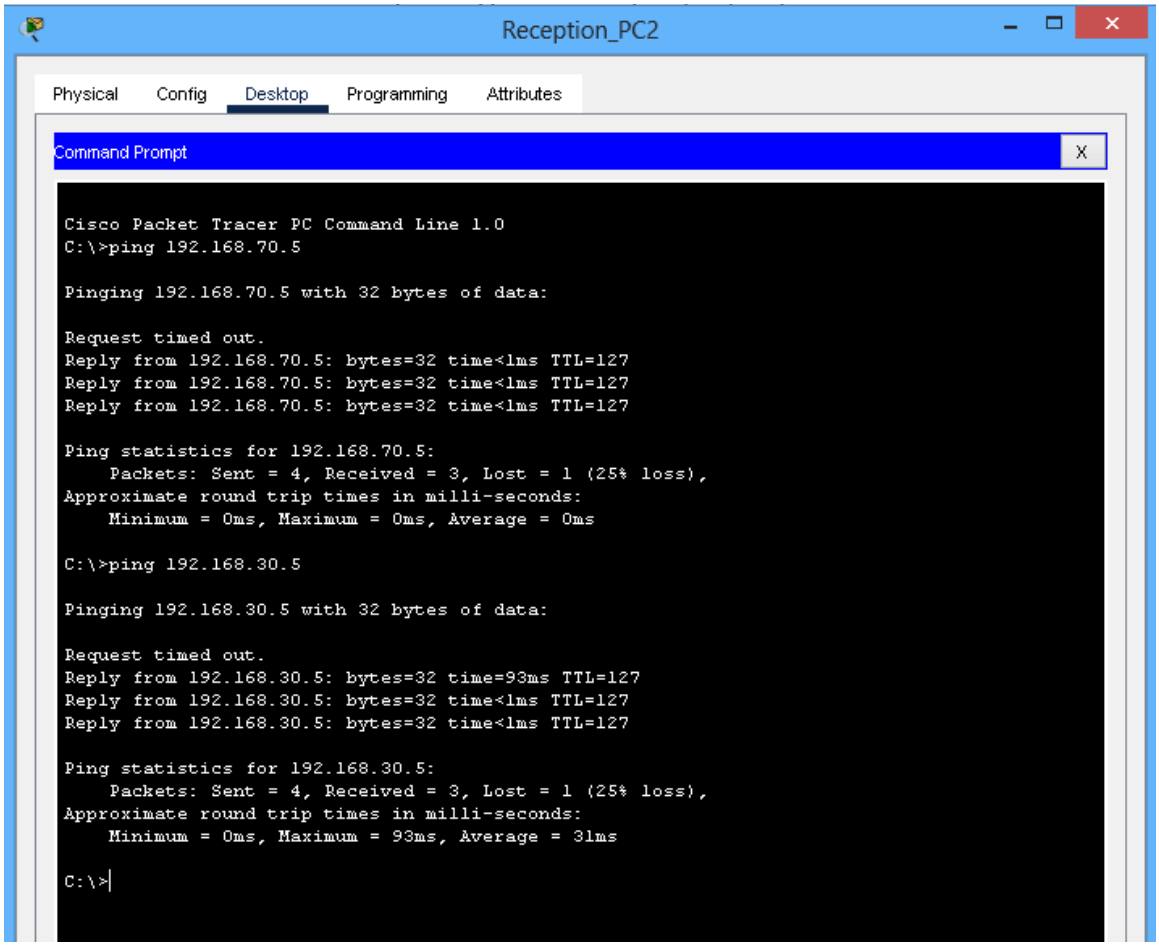
- PCs in each VLAN received correct DHCP addresses.

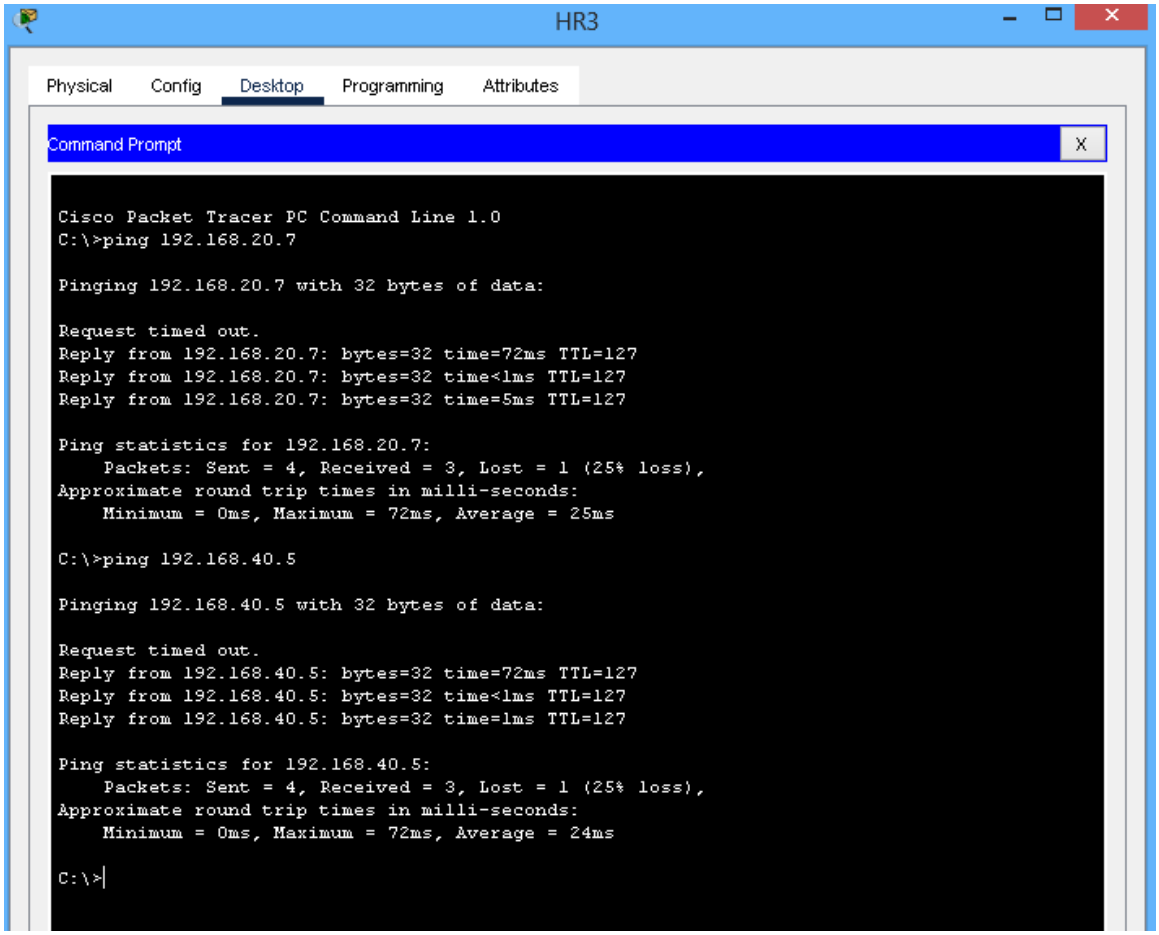






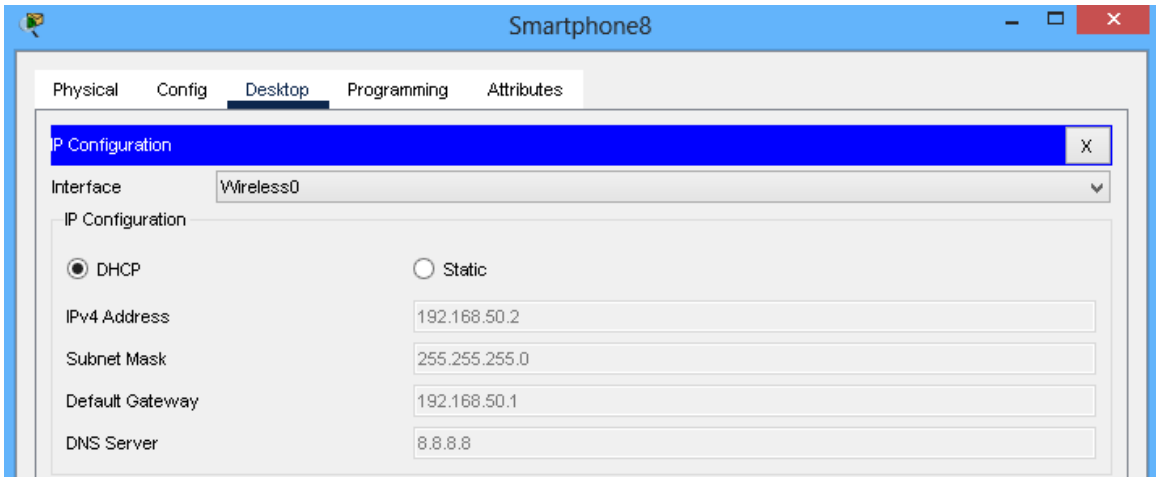
- PCs could ping their VLAN gateway.
- Devices in different VLANs could communicate via inter-VLAN routing



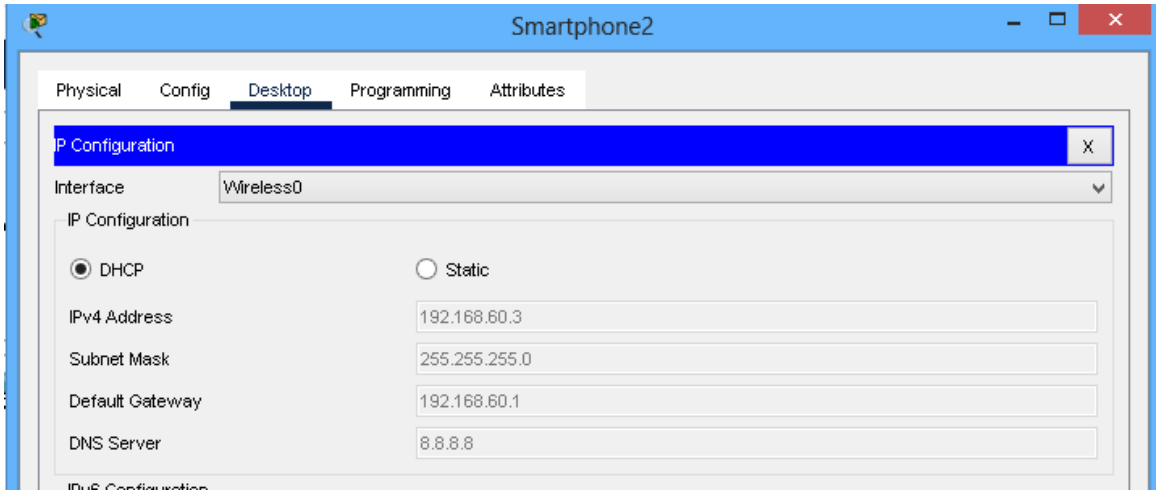


5.2 Wi-Fi Testing

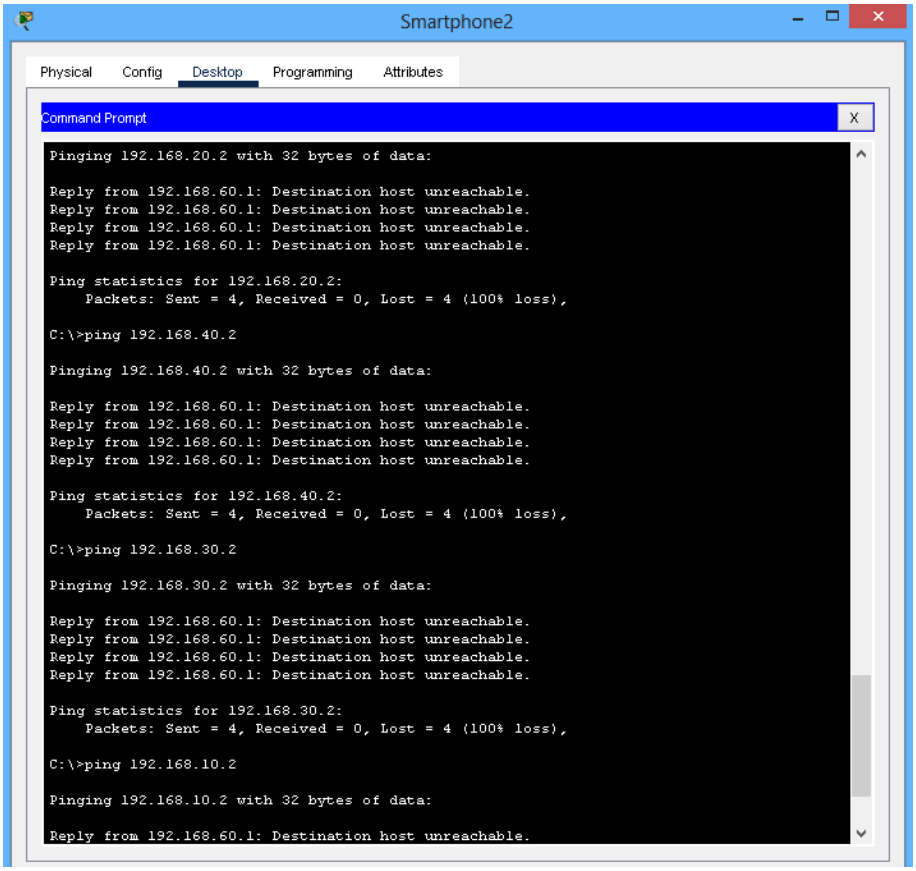
- Staff devices connected to Staff_WiFi (VLAN50)



- Guest devices connected to Guest_WiFi (VLAN60)



- Guest Wi-Fi could not access internal VLANs



6. Bill of Materials

Item	Qty	Unit Price (SAR)	Total (SAR)	Reason
Cisco 2911 Router	1	1,350	1,350	Used as the main core router to perform inter-VLAN routing between all hospital departments and to act as the default gateway for all VLANs.
Cisco Catalyst 2960 Switch	6	800	4,800	Used as access and distribution switches in all buildings to connect PCs, IP phones, printers, servers, and access points while supporting VLANs and trunking.
Access Point (802.11ac)	3	600	1,800	Provides wireless connectivity for staff and guest users in the waiting area, pharmacy, and employee café with good speed and coverage.
Servers (static IP)	7	3,300	9,900	Dedicated servers for DHCP services, Electronic Medical Records (EMR), and backup storage (NAS) to ensure reliable hospital data management.
PCs	78	1,500	117,000	Desktop computers used by doctors, nurses, administrative staff, laboratory technicians, and other hospital employees.
IP Phones	10	550	5,500	IP phones used in consultation rooms to support internal communication using the hospital network.
Network Printers	3	1,000	3,000	Shared printers used in reception, pharmacy, and administration to reduce cost and simplify printing management.
Network Cabling (UTP & Patch Cables)	1 set	3000	3000	Includes Ethernet patch cables and structured cabling required to physically connect all network devices across buildings.
Total			**159,550 SAR**	

7. Conclusion

This project successfully deployed a scalable, segmented hospital network using VLANs, Router-on-a-Stick inter-VLAN routing, and DHCP for automatic addressing. Separate Wi-Fi SSIDs provided secure access for staff and guests. All connectivity and security requirements were tested and verified.